

FUNCTIONAL DESIGN SPECIFICATION

Procure-to-Pay (P2P) & Record-to-Report (R2R)

Functional and Process Design Document for Accounting Software Development

Version	1.0
Purpose	Software Functional Design
Audience	Software Developers, Product Architects, Business Analysts, Functional Consultants
Scope	End-to-End Accounting & Bookkeeping System covering Procure-to-Pay (P2P) and Record-to-Report (R2R)

Document Objective

This document explains the accounting and bookkeeping processes required for designing an integrated accounting platform. The intention is to help software developers with no prior accounting knowledge understand:

- How accounting transactions originate
- How transactions flow through operational modules
- How journal entries are generated
- How transactions impact accounting records
- How financial statements are generated

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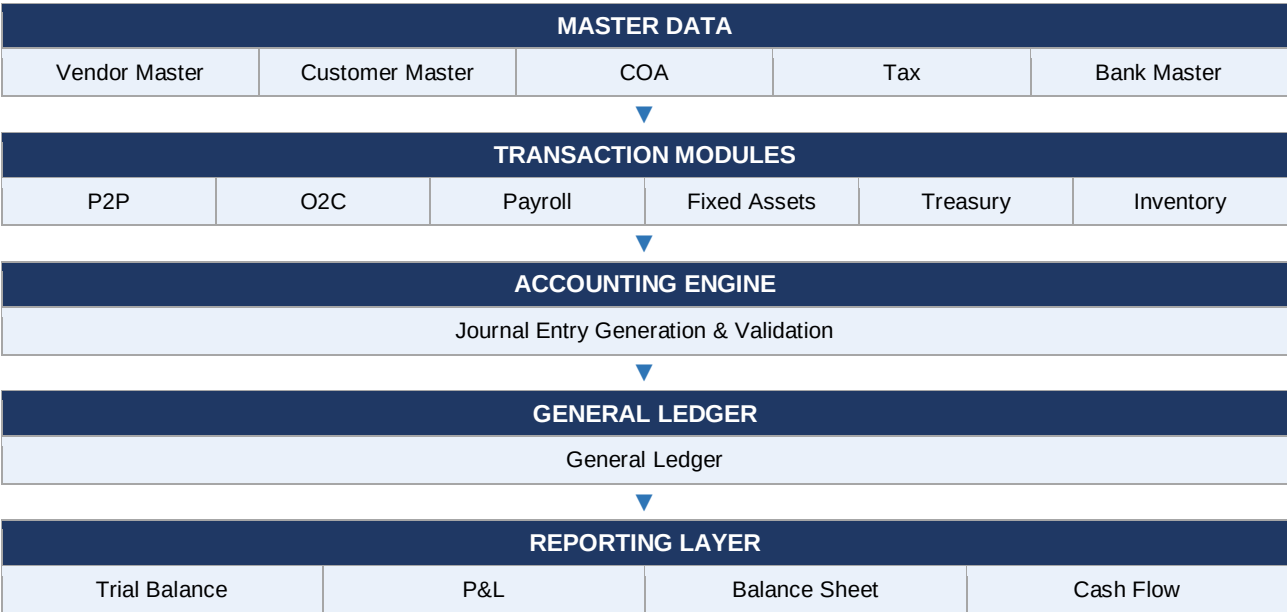
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1. Accounting System Overview

An enterprise accounting platform is usually built on two layers:

Operational Layer	Accounting Layer
Where business transactions originate. • Vendor onboarding • Purchase requests • Purchase orders • Invoice processing • Customer billing • Bank transactions	Where accounting entries are generated and posted. • Journal entries • General ledger postings • Trial balance • Financial statements

Overall Accounting Architecture (ERP Platform)

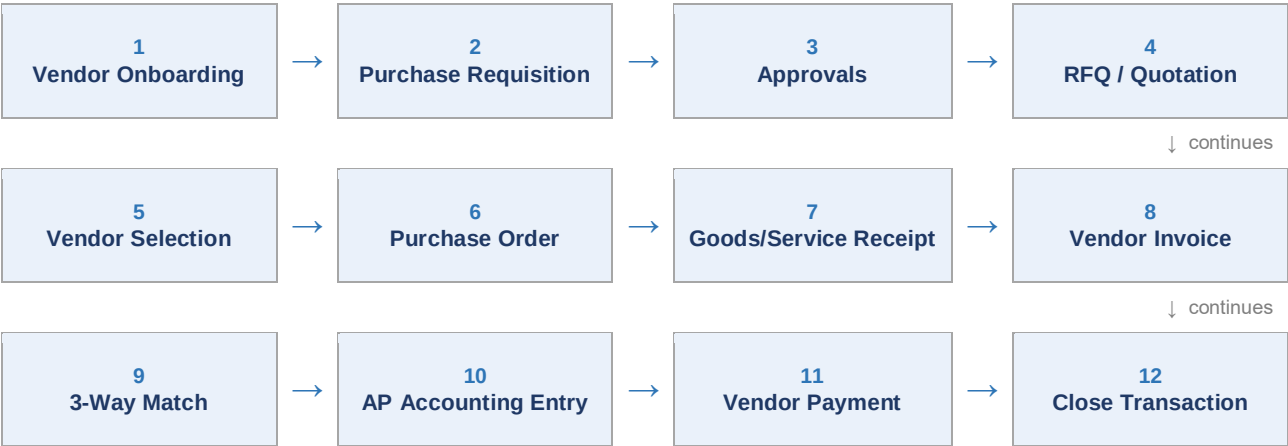


2. What Is Procure-to-Pay (P2P)?

Procure-to-Pay (P2P) is the complete process of:

- Identifying a requirement
- Procuring goods/services
- Receiving goods/services
- Recording supplier invoices
- Paying suppliers
- Posting accounting entries

P2P Lifecycle



3. Vendor Onboarding

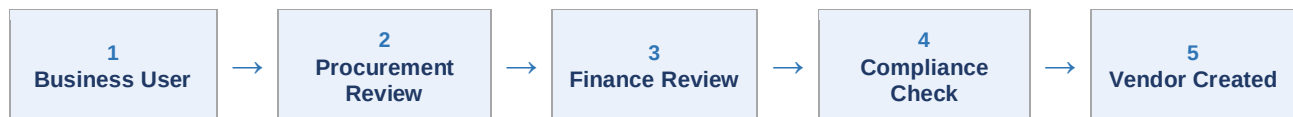
Objective

Create a supplier master record before any procurement activity can occur.

Information Required

Category	Data Elements
General Information	Vendor Legal Name; Trade Name; Address; Country; Business Registration Number
Tax Information	PAN; GSTIN; Tax Residency Certificate (where applicable)
Contact Information	Contact Person; Phone Number; Email
Banking Information	Bank Name; Beneficiary Name; Account Number; IFSC / SWIFT
Commercial Information	Payment Terms; Currency; Credit Period

Workflow



Vendor Master Database Structure

Vendor_ID	Vendor_Code	Vendor_Name	Country
GSTIN	PAN	Currency	Payment_Terms
Status	Created_By	Created_Date	

4. Purchase Requisition (PR)

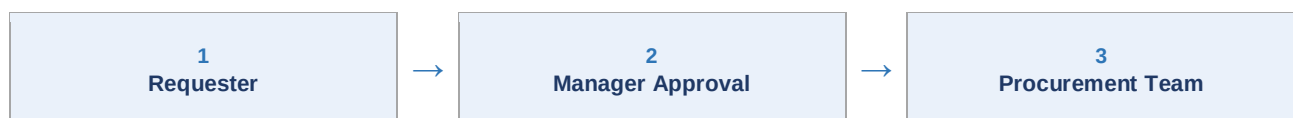
What Is a PR?

An internal request raised by an employee or department requesting goods or services.

Example

Marketing requests: 5 laptops; a printer; an annual software subscription.

PR Workflow



Core Fields

PR Number	Department	Cost Center
Requester	Item Description	Quantity
Estimated Cost	Need By Date	Business Justification

5. Request for Quotation (RFQ)

Purpose

Obtain quotations from multiple vendors.

Workflow



Example

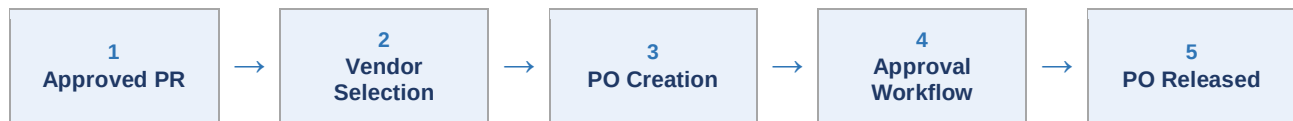
Vendor	Quoted Amount (₹)	Outcome
Vendor A	100,000	
Vendor B	95,000	Selected (lowest quote)
Vendor C	102,000	

6. Purchase Order (PO)

What Is a Purchase Order?

A legally binding purchase document issued to a vendor.

Purchase Order Flow



PO Information

PO Number	Vendor	Currency	Item
Quantity	Rate	Tax	Total Amount
Delivery Date	Payment Terms		

Approval Matrix Example

PO Value (₹)	Approver
0 – 50,000	Manager
50,001 – 500,000	Director
Above 500,000	CFO

7. Goods Receipt Note (GRN)

Purpose

Confirms physical receipt of goods. The supplier cannot usually be paid until goods are received.

Workflow



GRN Data

GRN Number	PO Number	Item	Received Quantity
Rejected Quantity	Receipt Date	Warehouse	

Example

PO Quantity	Received	Rejected
100	95	5

8. Service Entry Sheet (SES)

Used for service procurements. Examples:

- Audit Fees
- Consultancy Fees
- Legal Services
- IT Support

Workflow



9. Vendor Invoice Processing

Purpose

Record vendor's invoice in the accounting system.

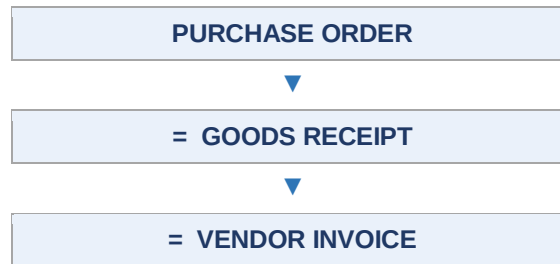
Invoice Fields

Invoice Number	Invoice Date	Vendor	PO Reference
Tax Amount	Invoice Amount	Currency	Due Date

10. Three-Way Matching

Critical Financial Control

The system should validate that the three documents agree:



Successful Match — Status: PASS

Document	Units
PO	100
GRN	100
Invoice	100

Failed Match — Status: EXCEPTION

Document	Units
PO	100
GRN	100
Invoice	120

Exception Workflow



11. Accounting Entry at Invoice Booking

Example

Material Cost = ₹100,000; GST = ₹18,000; Total Vendor Invoice = ₹118,000.

Journal Entry

Account	Debit	Credit
Inventory / Expense	100,000	
Input GST	18,000	
Accounts Payable		118,000

Accounting Impact

Asset Increases	Liability Increases
• Inventory • Input GST	• Vendor Payable

12. Accounts Payable Management

The AP team manages outstanding vendor balances.

AP Functions

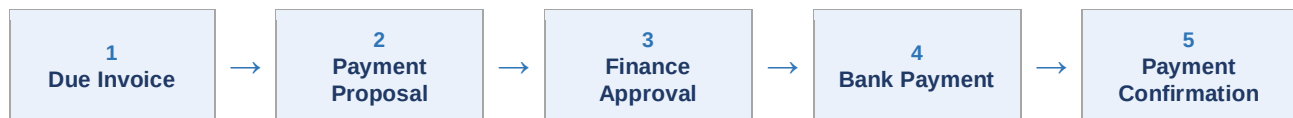
- Due date tracking
- Vendor ageing
- Hold invoices
- Payment proposals
- Reconciliations

AP Ageing Buckets

0–30 Days	31–60 Days	61–90 Days	90+ Days

13. Payment Processing

Payment Workflow



Payment Entry

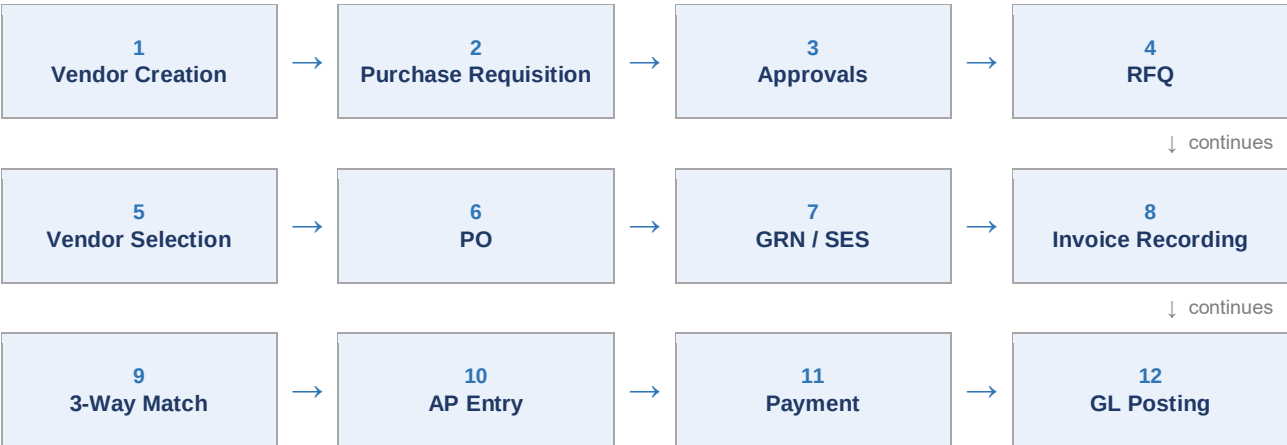
Invoice outstanding: ₹118,000.

Account	Debit	Credit
Accounts Payable	118,000	
Bank		118,000

P2P Completion

- PO Closed
- Invoice Closed
- Vendor Balance Cleared
- Accounting Updated

14. P2P Complete Process Map



15. What Is Record-to-Report (R2R)?

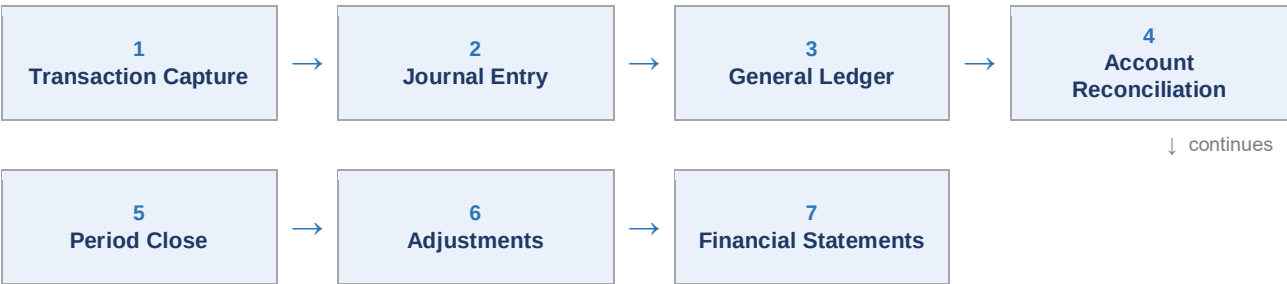
Record-to-Report (R2R) is the accounting cycle that collects all business transactions and converts them into financial statements.

Objective

Transform operational transactions into:

- Trial Balance
- Profit & Loss Account
- Balance Sheet
- Cash Flow Statement
- Management Reports

16. R2R Process Flow



17. Sources Feeding R2R

P2P	O2C	Payroll	Inventory	Fixed Assets
Banking	Treasury	Tax	Manual Journals	Intercompany

All transactions eventually create journal entries.

18. Chart of Accounts (COA)

The Chart of Accounts is the accounting structure used for posting transactions.

Sample COA

Code	Account	Type
100000	Assets	Group
110000	Bank	Asset
120000	Accounts Receivable	Asset
130000	Inventory	Asset
200000	Liabilities	Group
210000	Accounts Payable	Liability
220000	GST Payable	Liability
300000	Equity	Group
310000	Share Capital	Equity
400000	Revenue	Group
500000	Expenses	Group

19. Journal Entry Engine

Every accounting transaction must generate a balanced journal entry.

Rule

Total Debit = Total Credit

Journal Entry Structure

Journal Number	Posting Date	Entity	Account
Debit Amount	Credit Amount	Currency	Cost Center
Reference	Source Module		

Example

Account	Debit	Credit
Rent Expense	50,000	
Bank		50,000

20. General Ledger (GL)

The GL is the central accounting repository. All approved journals are posted to the General Ledger.

Posting Flow



21. Month-End Close Activities

Accruals

Expense incurred but invoice not received.

Account	Debit	Credit
Audit Fees Expense	100,000	
Accrued Liability		100,000

Prepayments

Expense paid in advance.

Initial Entry

Account	Debit	Credit
Prepaid Expense	120,000	
Bank		120,000

Monthly Recognition

Account	Debit	Credit
Rent Expense	10,000	
Prepaid Expense		10,000

Depreciation

Account	Debit	Credit
Depreciation Expense	Dr	
Accumulated Depreciation		Cr

Provisions

Account	Debit	Credit
Provision Expense	Dr	
Provision Liability		Cr

22. Reconciliations

Reconciliations ensure correctness of accounts.

Reconciliation	Compare	Against
Bank Reconciliation	Bank Ledger	Bank Statement
Vendor Reconciliation	Vendor Statement	Accounts Payable Ledger
GST Reconciliation	Purchase Register	GST Return Data

23. Trial Balance

A Trial Balance is generated after all postings.

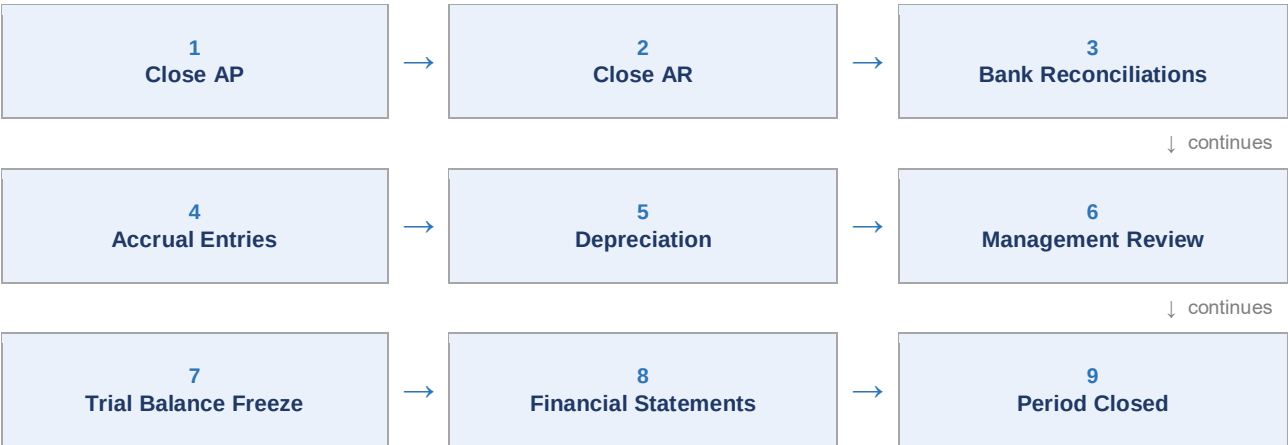
Sample Trial Balance

Account	Dr / Cr	Amount
Cash	Dr	1,000,000
Inventory	Dr	500,000
Accounts Payable	Cr	250,000
Revenue	Cr	5,000,000
Expenses	Dr	3,500,000

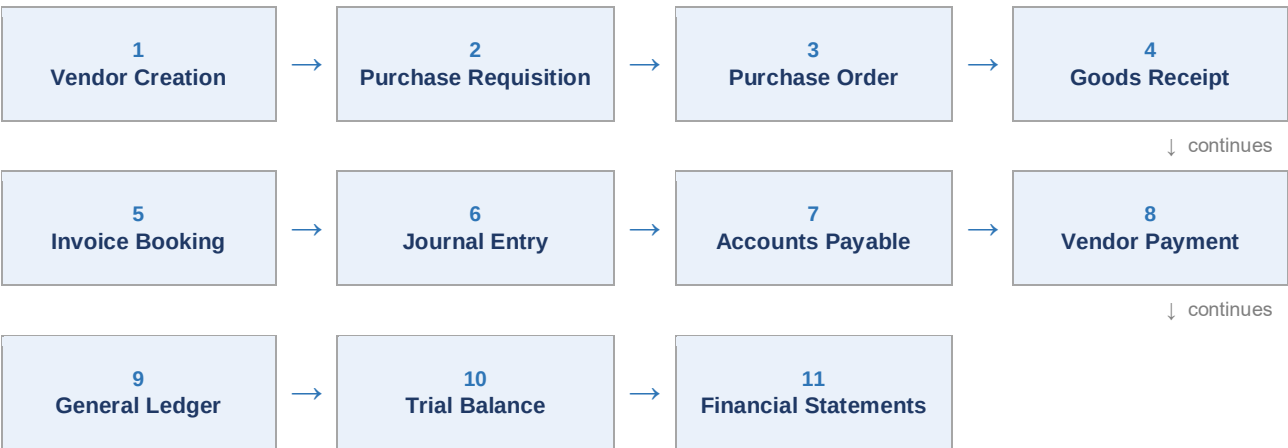
24. Financial Statements

Balance Sheet	Profit & Loss	Cash Flow Statement
Assets - Cash - Inventory - Receivables Liabilities - Payables - Loans Equity - Share Capital - Retained Earnings	Revenue Less: Expenses Profit Before Tax Tax Net Profit	- Operating Activities - Investing Activities - Financing Activities

25. Period Close Workflow



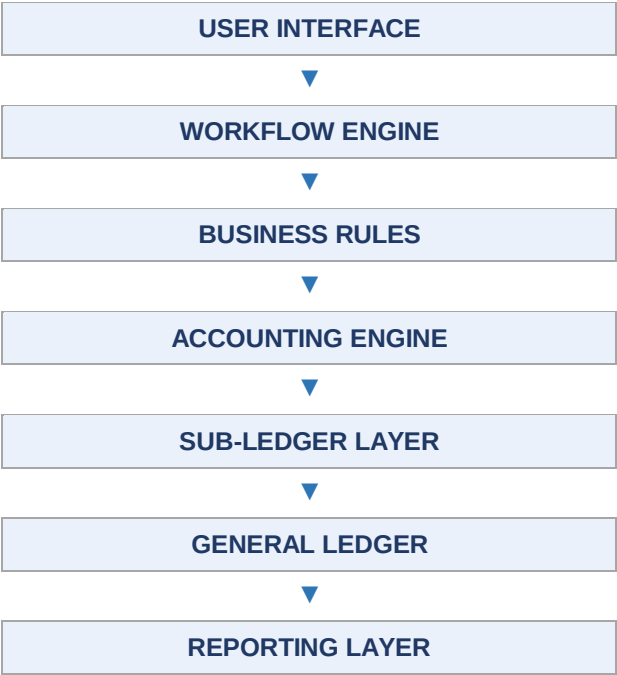
26. End-to-End Transaction Flow



27. Recommended Software Modules

Module Group	Components
Master Data Management	• Entity Master • Vendor Master • Customer Master • Employee Master • Bank Master • Currency Master • Tax Master • Chart of Accounts • Cost Center Master
P2P Module	• Vendor Onboarding • Purchase Requisition • RFQ Management • Vendor Evaluation • Purchase Order • Goods Receipt • Service Receipt • Invoice Processing • Accounts Payable • Payment Processing
R2R Module	• Journal Entry Management • General Ledger • Accruals • Prepayments • Fixed Assets • Reconciliations • Month-End Close • Consolidation • Financial Reporting
Workflow Engine	• Maker Checker Control • Multi-Level Approvals • Escalation Matrix • Email Notifications • Audit Logs
Reporting & Analytics	• Vendor Ageing • AP Ageing • Trial Balance • General Ledger • Balance Sheet • Profit & Loss • Cash Flow Statement • KPI Dashboard • Audit Reports

28. Recommended System Architecture



Critical Design Principle

Never generate financial statements directly from transaction tables.

Correct Flow



This architecture follows the design principles used by enterprise ERP solutions such as SAP, Oracle Fusion, Microsoft Dynamics 365, NetSuite, and other large-scale accounting platforms.

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